

RegLIC: Region-Adaptive Learned Image Compression

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Abstract

Learned image codecs usually apply one decoder depth to an entire image, although the benefit of additional synthesis varies across regions. We introduce **RegLIC**, a region-adaptive depth framework studied with the DCVC-UF intra codec. Its evaluated shared-exit path recovers one latent, executes later synthesis blocks only on surviving tiles, and assembles mixed-depth features through adapters and a common reconstruction head. Across 53 intra frames and five quality indices, learned routing saves 27.97% of modelled synthesis MACs at a nominal 0.1 dB YCbCr-MSE-ratio target under per-source calibration; 2.54 percentage points exceed ordered dithering, which mixes depths without content-dependent placement. A retrospective common delivered-loss cap preserves a routing margin, whereas a fixed-control QP32 replay does not yet resolve its paired advantage. We also define a complementary 2/4/6/8/10/12-block codec bank to study independent representation specialisation; its router is not part of the reported results. These controls distinguish savings from shallower decoding, spatial allocation, and the still-unmeasured effect on full-codec latency.

1. Introduction

The architecture of a learned codec fixes synthesis depth before it sees a particular image. In the DCVC-UF intra model [7], every spatial location traverses the same twelve synthesis blocks, although additional blocks need not improve every location equally. This uniform schedule makes decoder capacity easy to provision, but ties the cost of an entire frame to the deepest path. We ask a narrower question than whether the decoder can be made smaller: *can its existing capacity be spent where it improves a particular decoded image?*

Spatially adaptive computation has clear precedents. ClassSR sends super-resolution patches to networks of different sizes [6]; APE instead predicts when a patch should leave a shared restoration network [12]. In compression, AdaNIC already routes spatial regions through dynamic transforms [11]. These works motivate allocation, but they do not an-

SOURCE LUMA

EXECUTED DEPTH

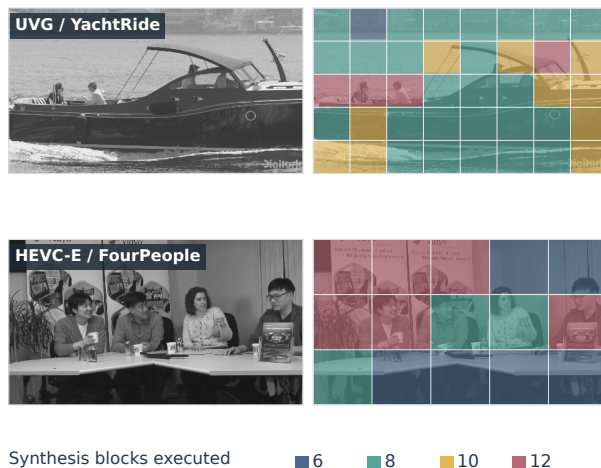


Figure 1. **Measured spatial allocation, rather than a conceptual scene.** Source luma and the archived QP32 shared-exit maps for two images from a predeclared eight-source gallery. Colours encode the number of synthesis blocks actually executed by each tile (6/8/10/12). These source-calibrated maps illustrate nonuniform computation; the luma images are inputs, not reconstructions, and the maps alone do not establish a visual-quality gain.

swer the decoder-depth question in DCVC-UF. The decoder receives a *coded* latent: texture visible in the source is useful only if the bitstream preserves information that further synthesis can recover. Moreover, the chosen path must truly skip computation while returning tiles to one coherent image. An exit selected after evaluating every block would have no decoding benefit.

We study two distinct ways to expose this capacity axis under the name **RegLIC** (Figs. 2 and 3). The evaluated system shares one analysis transform, bitstream and decoder prefix. A tile continues through conditional block pairs only while it remains active; an exit adapter maps its features to a common space, and a full-frame repair and reconstruction head assembles the result. A complementary bank of inde-

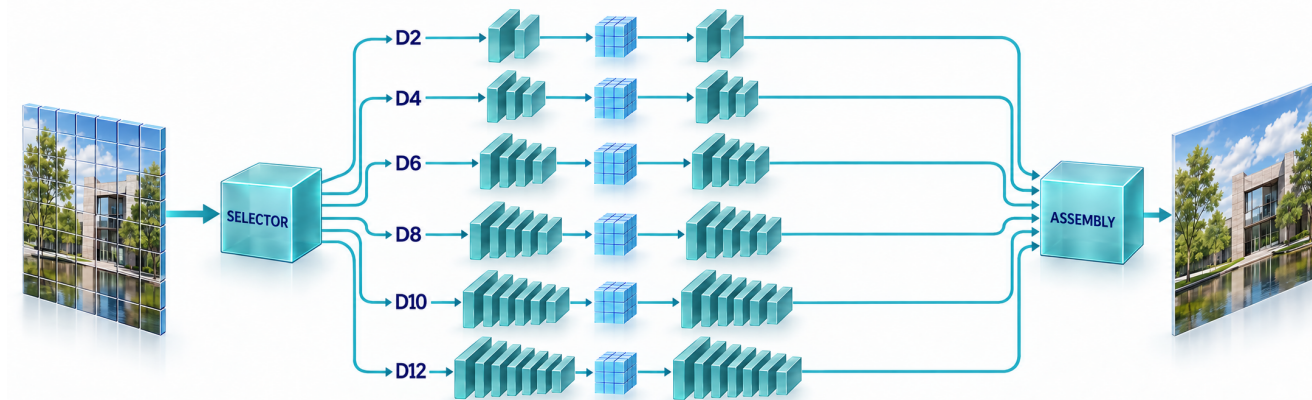


Figure 2. **Independent-depth codec bank.** A selector assigns each source region to *one* of six complete DCVC-UF codecs before encoding; each path has independent analysis, entropy and synthesis weights. Routes depict alternatives, not six executions per region. Decoded regions are assembled into an image. The six-way selector is conceptual and has no reported measurements; it is separate from the shared early-exit path in Fig. 3.

pendently trained 2/4/6/8/10/12-block codecs would route *before* encoding, allowing the representation and entropy model to specialise with depth. That bank changes the rate and boundary problem as well as the compute budget. We therefore keep its training study separate from the measured shared-exit results and do not claim six-way routing performance here.

The crucial comparison is between *using fewer blocks* and *placing* the remaining blocks well. Across 53 intra frames and five quality indices, source-calibrated shared-exit routing saves 27.97% of modelled synthesis MACs at a nominal 0.1 dB YCbCr-MSE-ratio target. A non-content-dependent depth mixture accounts for much of this saving; the extra benefit of learned placement is smaller and varies with the quality budget. On the QP32 fixed-control replay, that paired benefit is not yet resolved. These results support conditional execution and measured arithmetic savings, while leaving source-independent routing gains and full-codec latency as separate questions.

Overall, our contributions are three-fold:

1. We formulate spatial synthesis-depth allocation for DCVC-UF and distinguish a shared-latent exit from a bank of independently coded depths, making their different rate and boundary costs explicit.
2. We realise genuine tile-wise early exit: inactive tiles skip later block pairs, and depth-specific adapters feed a common full-frame repair and reconstruction path.
3. We evaluate the executed-depth/quality trade-off against uniform and non-content-dependent allocations, with calibration controls that separate arithmetic savings from the still-unresolved placement gain.

2. Related Work

Reducing codec complexity. Shallow synthesis trades decoder work for stronger analysis or iterative encoding [15]. DCVC-RT attacks memory traffic and function-call overhead [4]; DCVC-UF streamlines entropy interaction [7]. Our intra extension instead asks which regions merit additional synthesis. Modelled MACs and complete execution time are reported separately.

Spatial network selection and early exit. ClassSR assigns restoration patches to networks of different capacity [6]; APE predicts the incremental value of later layers in a shared multi-exit network [12]. AdaRevD routes deblurring patches [8], and RBQE terminates enhancement early [13]. Swift provides compute-dependent exits in layered video decoding, at different output resolutions [2]. RegLIC’s evaluated path retains one entropy-decoded latent and a fixed output resolution while processing only surviving tiles through later blocks.

Adaptive learned compression. Slimmable autoencoders optimise multiple widths [14]; ELFIC selects per-instance decode complexity [17]; ABC adapts structure to content and budget [16]. AdaNIC routes spatial transform capacity [11], and cgSlimDecoder allocates video-decoder channels [3]. MixCompress selects depth experts using rate embeddings rather than image content [10]. MoECodec routes content-dependent token experts and regularises their spatial coherence [18]. RegLIC instead varies constant-width synthesis depth after a shared entropy-decoded latent and compares learned placement against blind mixtures with the same available depths.

Independent codecs and region boundaries. Blard et al. select among content-specialised complete codecs and jointly process connected same-mode regions [1]. Our prospective bank instead varies synthesis depth and chooses a codec before encoding; shared exit holds the representation fixed. Boundary context matters in either case. Kim et al. preserve full-image CNN reconstruction using overlap rules [5]; our zero-halo conditional suffix accepts context loss and learns full-frame repair, so tiled and full-frame controls remain distinct.

3. Spatially Adaptive DCVC-UF

3.1. The allocation problem

Let $\mathbf{a} = (a_1, \dots, a_N)$ assign a synthesis option to each region. We seek the least computation that keeps the assembled reconstruction within a quality-loss budget:

$$\min_{\mathbf{a}} \mathcal{C}(\mathbf{a}) \quad \text{subject to} \quad \Delta(x, \hat{x}_{\mathbf{a}}; \hat{x}_{\text{ref}}) \leq \delta, \quad (1)$$

where $\mathcal{C}$ is the chosen compute cost and $\hat{x}_{\text{ref}}$ fixes the reconstruction anchor. The main sweep uses $\Delta_{444} = 10 \log_{10}[D_{444}(x, \hat{x}_{\mathbf{a}})/D_{444}(x, \hat{x}_{\text{ref}})]$, with $D_{444} = \|x - \hat{x}\|_2^2/(3HW)$ on equal-channel, unclipped YCbCr 4:4:4 tensors. Positive values denote a quality loss in dB. The constraint belongs to the *assembled image*: spatial interactions can invalidate an additive tile-error approximation.

For early exit, a_t identifies a stopping point after one latent has been recovered. For the prospective model bank, a_t identifies a complete codec before encoding, making both payload rate and reconstruction depend on the assignment. The bank must therefore also be compared at matched rate, including expert signalling (Sec. 5). We implement shared exits below.

3.2. Shared latent, variable depth

Let $\hat{y}$ be the recovered latent of the fixed DCVC-UF analysis transform and entropy model. The synthesis network upsamples it by two and processes 384-channel features at one eighth of the image resolution. Four of its twelve trunk blocks form the full-frame stem; the remaining eight form four conditional block pairs.

The stem output is partitioned into 32×32 feature tiles, each covering 256×256 image pixels. For a tile assigned depth k , let $f_t^{(k)}$ be its feature after the corresponding block pair and let A_k be its adapter. The complete image is reconstructed as

$$\hat{x}_{\mathbf{k}} = H\left(q \odot G\left[\text{stitch}_t A_{k_t}(f_t^{(k_t)})\right]\right), \quad (2)$$

where G is grid-conditioned repair, q is the decoder’s quality scaling, and H is the common output head, including pixel shuffle. Adaptation acts on features before the shared head, so tiles with different stopping depths need not be stitched directly in image space.

Reachable depth and actual execution. The reachable depths are $\mathcal{K} = \{6, 8, 10, 12\}$; depths two and four are inside the shared stem and are not evaluated exits. At each conditional stage the implementation gathers only the surviving tiles, executes the next block pair, and writes departing tiles through their adapters. A deeper tile reuses all computation already performed. Evaluating all four candidates and masking their reconstructions would not implement this execution saving. For the recorded router map at 0.1 dB, the active population decreases from 40 to 27, seven and three tiles (Fig. 4).

3.3. Exit alignment and boundary context

An intermediate feature is not automatically compatible with the final head. The exit adapter supplies the missing transformation with pointwise residual operators. Its residual output is zero-initialised, preserving the inherited feature at initialisation. Adapter capacity scales with the depth replaced. Pointwise operators avoid adding an exit-specific spatial receptive field; their arithmetic is included in the cost model. The deepest exit uses an identity connection rather than an additional learned adapter.

Independent tile execution still removes context from the trunk’s spatial operators. The conditional trunk uses replicate padding without a halo. After stitching, a depthwise spatial filter and pointwise mixing act on the full feature canvas. A learned gate indexed by position within the tile gives the repair access to the known boundary lattice. The head then operates on the complete canvas. This design places context recovery at the output stage rather than expanding every conditional block with an overlapping halo. We measure the trained checkpoint’s dependence on adapters and repair in Sec. 4.7; matched retraining is needed to choose their capacity.

3.4. Learning exits and their selector

The evaluated shared-latent checkpoint, denoted e15, is initialised from released DCVC-UF weights. Analysis and entropy parameters remain frozen; the synthesis network, adapters and repair are fine-tuned. Training samples tile depths and applies reconstruction supervision through the mixed tiled path. Thus the adapters learn under the same stitching and full-frame head used at inference. The inspected training configuration also enables a full-depth reconstruction anchor against the frozen release and adjacent-exit feature distillation. Its objective can be written as

$$\mathcal{L} = \mathbb{E}_{\mathbf{k}}[R + \lambda D(x, \hat{x}_{\mathbf{k}})] + a\mathcal{L}_{\text{anchor}} + b\mathcal{L}_{\text{feature}}. \quad (3)$$

Here the latent-rate term R is shared across exits. With the front end frozen it does not optimise exit-dependent rate. The archived fine-tuning configuration uses 512-pixel crops and batch size eight; it is distinct from the new full 105-

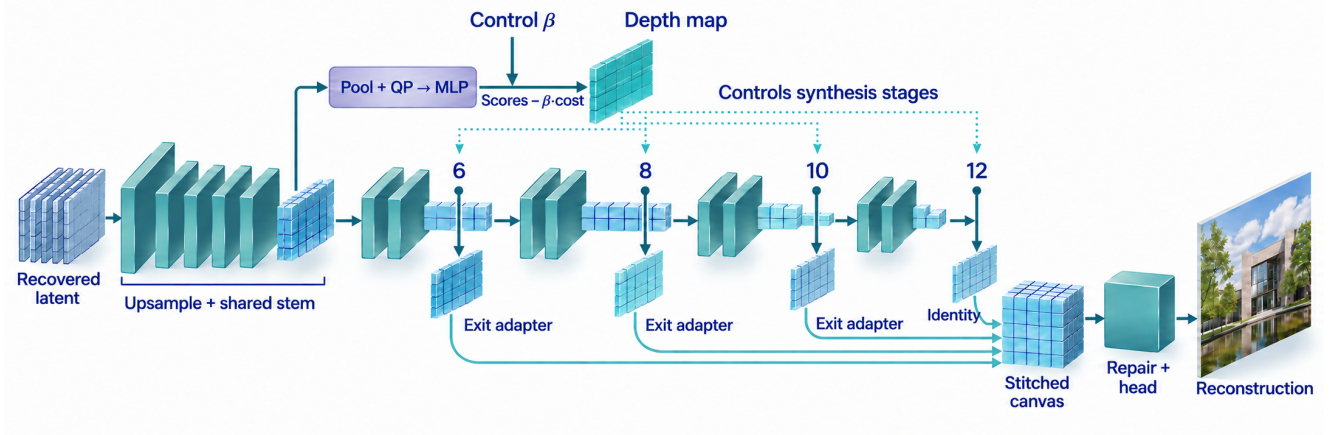
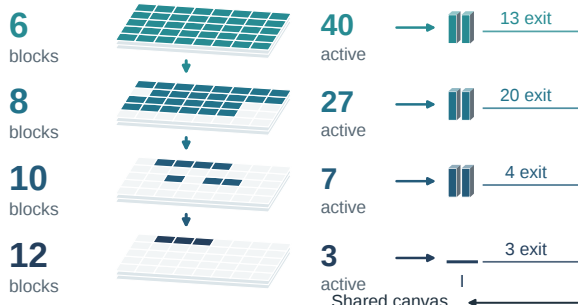


Figure 3. **Share the representation; spend further depth only on surviving tiles.** A four-block shared stem supplies the MLP with features and QP. Cost-adjusted scores select exits at total depths 6/8/10/12. Each departing feature passes through its adapter (identity at depth 12) into one canvas, followed by full-frame repair and reconstruction. The illustration follows the implemented connectivity; tensor contents and the output scene are conceptual, AI-generated artwork rather than measured reconstructions.

Spatial work disappears as tiles exit

Recorded 5×8 tile grid · one shared latent



77 / 160 tile-block-pair evaluations

Counts describe execution; they are not timings.

Figure 4. **A spatial decision becomes a smaller workload.** The recorded videoSRC05 map at QP32 leaves fewer tiles active at each successive block pair. Coloured cells run the indicated stage at their padded spatial coordinates; pale cells have already exited. Departures feed the aligned canvas in Fig. 3. The suffix evaluates 77 tile pairs versus 160 for an all-deep map; this work count does not measure GPU time.

epoch official-recipe depth study in Sec. 5. The supplement records the available training provenance and its limits.

The stopping predictor projects the stem from 384 to 48 channels, pools each tile’s mean and standard deviation, and incorporates normalised QP. An MLP with two 256-unit hidden layers produces exit log scores. For candidate tile errors $M_{t,k}$ and normalised costs c_k , its soft training labels favour low values of $M_{t,k} + \lambda_c c_k$. Soft-label cross-entropy and a regret term penalise costly decisions while allowing nearly equivalent exits to receive similar scores.

The checkpoint uses $\lambda_c = 1.3 \times 10^{-5}$ and temperature ratio 0.1; the exact target, input layout and optimisation settings are given in the supplement. This objective learns a preference among exits, not a bound on final image error.

3.5. Stopping under a quality target

For equal padded tiles, the arithmetic model is

$$C(\mathbf{k}) = \frac{1}{N} \sum_t c_{k_t}, \quad S(\mathbf{k}) = 100[1 - C(\mathbf{k})]. \quad (4)$$

Shared stem and output work, exit adapters and conditional depth are included in c_k . Entropy-recovery networks, arithmetic coding, router execution, signalling and system overhead are outside this synthesis MAC quantity. The four costs are approximately $\{0.609, 0.758, 0.870, 1.010\}$ relative to full-frame synthesis. In particular, a tiled all-deep map can cost more than the full-frame reference because repair is still executed.

Source-informed search minimises $M_{t,k} + \lambda_c c_k$ per tile and adjusts λ for a target table loss. The learned alternative uses

$$k_t(\beta) = \arg \max_k \{\ell_{t,k} - \beta c_k\}, \quad (5)$$

where ℓ denotes log scores. Uniform depth selects one exit per frame. Bayer dithering mixes adjacent exits with a fixed spatial ordering. Its mixture fraction is calibrated under the same target, so it controls for finer average depth without predicting spatial difficulty.

In the recorded sweep, all controls are calibrated per source frame using stored error tables. This measures the allocation available with encoder-side source access. The fixed-control experiment in Sec. 4.6 removes per-source calibration. Increasing β cannot increase selected cost, but

source distortion need not be monotone; the supplement audits the finite assignment path for both router and dither.

3.6. Quality references and decision information

The table target compares padded YCbCr 4:4:4 candidate errors to a released-weight full-depth reference. Final reported loss instead compares the cropped mixed output to the cropped, fine-tuned e15 full-frame output. These anchors are not interchangeable: the encoder/entropy weights match, while inherited synthesis weights have changed. Full-frame repair also couples neighbouring tiles. A table threshold therefore does not guarantee the same numerical loss after final reconstruction. We retain this distinction in the nominal sweep and add a retrospective common-reference delivered-loss comparison. A separate QP32 replay on all 53 frames measures the e15 full-frame RGB loss against released synthesis as 0.00908 dB; details are in the supplement.

An encoder-selected map must be signalled to the decoder; alternatively, a decoder-local predictor must recover the same decisions from available features and a reproducible control. We use explicit two-bit exit indices in a CPU stream-only check (Sec. 4.7). Its common payload confirms the representation-sharing path, while headers and map bytes remain part of the complete rate. Source access is legitimate at the encoder; its candidate evaluations and calibration must be charged to the decision cost.

4. Experiments

4.1. Evaluation protocol

We evaluate the first intra frame of 53 sequences at QPs $\{0, 16, 32, 48, 63\}$: 265 frame-QP pairs, six targets from 0.05 to 0.5 dB, and identical e15 weights, tile geometry and exit costs for all policies. The archive contains actual mixed reconstructions. Additional CPU replays test fixed controls, reference drift and component dependence. All decoder comparisons concern DCVC-UF. An exploratory one-frame, synthesis-only A6000 measurement appears in the supplement; complete-codec GPU latency remains unmeasured.

Two complementary protocols answer different questions. The *nominal sweep* compares maps selected under the same table target, retaining the common feasible cases of all four rules: 204 pairs at 0.05 dB, 263 at 0.1 dB and 265 at each larger target. The *delivered-cap analysis* reselects from the outputs already measured at those targets under a common cropped YCbCr 4:4:4-loss cap, with the full-frame e15 output as an explicit fallback. It retains all 265 pairs at every cap. The second protocol is retrospective and source-aware; its cost omits acquiring and rejecting candidates.

Confidence intervals use 5,000 paired bootstrap draws over sequences, keeping QPs from the same sequence together. They quantify sequence-sample variation conditional on fixed weights and controls, without including training

or calibration uncertainty. The supplement also reports a conservative content-group sensitivity check. A separate fixed-cohort comparison retains the same 204 pairs across the entire nominal sweep, avoiding changes of sample membership as the target relaxes.

4.2. The reconstruction value of depth

The stored uniform-exit reconstructions provide a direct depth control. Across all 265 pairs, their mean padded YCbCr 4:4:4 losses against the released full-frame reference are 0.205/0.097/0.061/0.038 dB at 6/8/10/12 blocks. Their modelled MAC savings are 39.12/24.22/13.03/−0.95%, respectively. Additional depth has diminishing average benefit and non-uniform local benefit: mean within-frame tile-gain IQRs are 0.075/0.034/0.024 dB for successive block pairs. The supplement gives the distributions and paired intervals. These measurements concern actual uniform maps on padded YCbCr 4:4:4; they motivate spatial allocation but are not cropped mixed-map losses or results for independently trained shallow codecs.

4.3. Early exit reduces work; routing improves its allocation

At the nominal 0.1 dB target, RegLIC with learned routing saves 27.97% of modelled synthesis MACs. Source-informed search reaches 29.52%, while uniform depth and dithering reach 21.07% and 25.43%, respectively (Fig. 5). The ordering separates three effects. Shallower synthesis supplies most of the reduction. Mixing adjacent depths improves the available cost resolution. Content-dependent placement contributes a further 2.54 percentage points over dithering, with a paired 95% interval of [1.83,3.26].

4.4. The gain persists under a common delivered-loss cap

Does the margin persist when every selected output meets the same measured loss cap? For each policy and frame-QP pair, we deduplicate the maps measured at the six nominal targets, retain those meeting the cropped Δ_{444} cap within 10^{-4} dB, and choose the cheapest. Repeated maps have identical recorded losses and costs. The evaluated e15 full-frame output supplies a zero-loss, zero-saving fallback; no quality point is interpolated.

Figure 6 reports the 0.1 dB case. The router saves 27.75% versus 24.81% for dithering, a 2.93-point paired difference [2.11,3.79]. Their mean achieved losses are also close: 0.0786 and 0.0791 dB. Only one router case uses the full-frame fallback, compared with two dither cases. The margin survives a common measured cap. Selection remains retrospective and source-aware, omits candidate-acquisition cost, and searches only the recorded outputs.

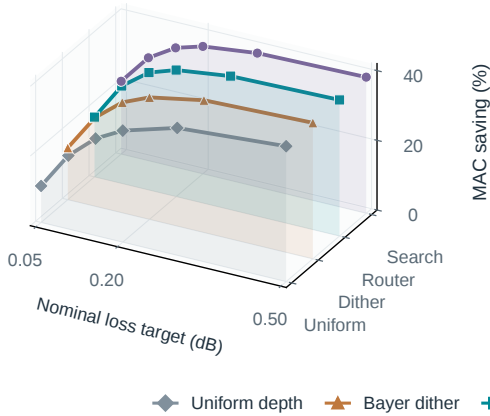
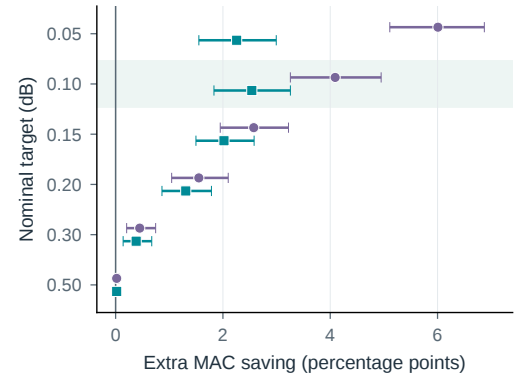
a Budget shapes the compute frontier**b** Increment over dithering

Figure 5. **Early exit supplies the saving; spatial routing adds a budget-dependent margin.** **a**, Measured operating points shown in orthographic 3D; the policy axis is categorical and ribbons only guide the eye. **b**, Paired advantages over Bayer dithering, with 95% sequence-cluster intervals; the shaded row highlights 0.1 dB. QP and fixed-cohort analyses are in the supplement. All curves use per-source calibration. MAC savings exclude router and signalling overhead and are not measured runtime reductions. Targets use the Δ_{444} MSE-ratio loss defined in Sec. 3.

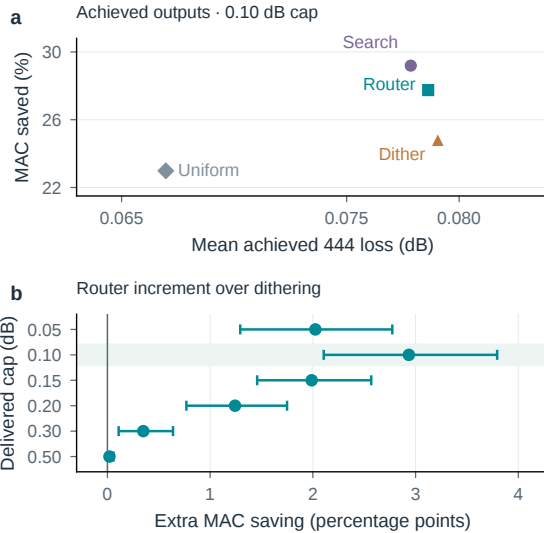


Figure 6. **Placement matters most before the cheap exit saturates.** **a**, Achieved mean loss and MAC saving for all four policies at the 0.1 dB cap. **b**, Paired router–dither saving differences across measured YCbCr 4:4:4-loss caps, with 95% sequence-cluster intervals. All 265 pairs remain; uniform/dither/router/search use the e15 full-frame fallback 11/2/1/2 times. Selection is retrospective and source-aware.

4.5. Why the routing margin contracts

The cheapest reachable depth imposes a 39.12% ceiling on the arithmetic model. As the target relaxes, more tiles

can take that exit and there are fewer consequential spatial decisions. The nominal router–dither margin declines from 2.54 points at 0.1 dB to 1.31 at 0.2 dB, 0.38 at 0.3 dB and 0.02 at 0.5 dB. Under the delivered-cap protocol the margin similarly falls from 2.93 to 0.35 points between 0.1 and 0.3 dB. The fixed-cohort result and QP breakdown in the supplement indicate that this behaviour is not explained solely by a changing feasible subset. At the 0.1 dB delivered cap, routing saves more in 170 pairs, less in 24, and ties in 71. At 0.5 dB, 262 of 265 pairs tie; the residual gain comes from one sequence.

The eight sources in Fig. 7 show how routing changes with content and budget. Relaxing the target admits more shallow exits, while routing, dithering and source search spend depth differently. A fixed sequence-name hash and dataset quotas select sources without using outcomes; infeasible cases remain. Luma supplies spatial context, not quality evidence.

A reconstruction intervention isolates placement from exit proportions. For every QP32 Q90 map, we replay three permutations within equal valid-tile extents, preserving depth counts and depth-weighted valid area: 53 frames and two policies give 424 map cases. Relative to the mean shuffled RGB MSE, original router placement gains 0.0139 dB [0.0095,0.0193]; dithering’s gain is 0.0004 dB [−0.0003,0.0010]. Unchanged maps remain included. These intervals condition on three sampled permutations and the fixed development cohort. The intervention supports spatial value, not a net-runtime or external-test advantage.

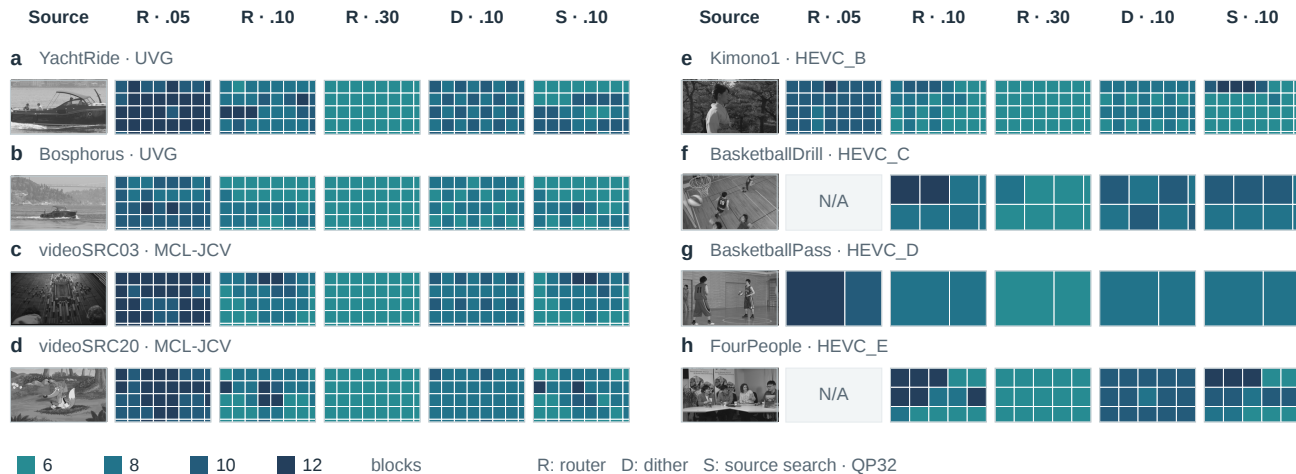


Figure 7. **Adaptive computation across content and budget.** a–h, Eight sources selected by sequence-name hash within fixed dataset quotas. Each row shows source luma, router maps at nominal 0.05/0.10/0.30 dB, and dithering/source-search maps at 0.10 dB. Colours encode executed depth on valid image extent; N/A retains infeasible cases. All maps are recorded at QP32 with source-calibrated controls. Source thumbnails show content, not reconstruction quality.

4.6. Fixed controls test the routing advantage

To remove per-source calibration, we fit each scalar control on four sequence folds and apply it to the fifth, keeping all QPs together. Calibration minimises compute under a mean or 90th-percentile (Q90) table-loss target. Actual QP32 replay gives 318 reconstructions: 53 frames, three policies and two controls, with no quality fallback.

Figure 8 reports cropped RGB loss against e15 full-frame synthesis. Q90 calibration lowers the observed frequency of losses above 0.1 dB, at the cost of less arithmetic saving. The router–dither saving difference is 1.87 points under mean control, with a paired 95% interval of $[-1.60, 5.19]$, and 2.54 points under Q90, with interval $[-0.42, 5.54]$. Both include zero, and the policies do not have exactly matched achieved quality. Thus the source-calibrated margin has not yet been established for fixed controls.

This is a calibration diagnostic on development-used data, not an untouched test set. Two similar-content pairs cross the sequence folds; grouping them in a bootstrap sensitivity check leaves both intervals crossing zero. An aligned released-weight replay also measures the reference drift: e15 full-frame loses 0.00908 dB mean RGB PSNR against released D12. Using released D12 raises absolute policy losses by that same per-image offset without changing their paired differences. The supplement gives the split, predictor and metric audits.

Fitting β on 24 DIV2K crops, then freezing it, gives 20.05% analytical synthesis MAC saving at 0.0889 dB cropped Δ_{444} on 24 disjoint DIV2K crops and 25.88% at 0.0976 dB on Kodak24 $\times$ QP5. The Kodak mean hides 47/120 losses above 0.1 dB and QP48/63 means of

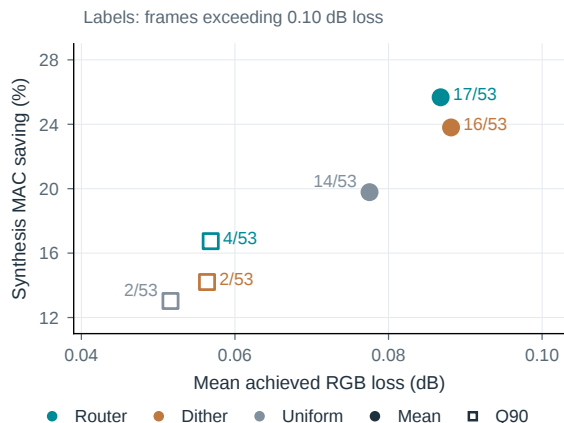


Figure 8. **Fixed controls trade saving against quality risk.** Actual reconstructions on all 53 QP32 frames. Colour identifies the policy; filled circles and open squares denote mean and Q90 calibration. Labels count RGB losses above 0.1 dB. Axes report achieved RGB loss against e15 full-frame and modelled synthesis MAC saving. Controls fitted on other sequence folds do not match achieved quality.

0.1275/0.1261 dB. The supplement reports the full transfer protocol, uncertainty and a more aggressive calibration rule that fails transfer.

4.7. Checking the mechanism

Exit alignment and repair. With the checkpoint and 53 Q90 maps fixed, we replace adapters, repair, or both by identity and retain all images. Adapter removal costs

1.2309 dB mean RGB PSNR, with a paired 95% interval of [0.8765, 1.6582]. Repair removal costs only 0.00208 dB [0.00060, 0.00363]; removing both costs 1.2538 dB. The trained model depends strongly on exit alignment. The small repair benefit motivates a matched-retraining and timing comparison before increasing its capacity. These interventions measure dependence of fixed weights, not the best performance attainable without each module.

Stream-only reconstruction. A separate CPU decoder receives the common entropy payload and explicit two-bit exit indices. On two predeclared QP32 frames with four maps each, all eight images, latents, symbols and entropy indexes match encoder-side reconstruction exactly; inner entropy bytes stay identical across maps. This verifies the shared-payload path on those cases. Map/header bytes and malformed-stream checks are in the supplement; native CUDA interoperability and full-cohort timing remain unmeasured.

4.8. What the encoder pays for a stopping decision

Source-informed search needs candidate reconstructions to populate its error table. The archived QP32 benchmark records 1,023.26 ms for four distinct candidate outputs and 336.87 ms for a shared tiled pass. These upper medians pool 24 trials over eight frames; raw timing distributions and device load were not retained. They indicate the cost of acquiring decision information, not an end-to-end speedup. The supplement details the stages.

A fixed-control predictor can avoid the error table; the source-calibrated router cannot, because its per-frame β uses the same information. Latency comparisons must charge feature generation, calibration, signalling, conditional execution and verification. Where learned and blind maps nearly coincide, even small prediction overhead can erase the benefit.

5. From Shared Exits to Independent Depths

Shared early exit uses one coded representation. Our complementary 2/4/6/8/10/12-block DCVC-UF bank instead allows each codec to learn its own analysis and entropy model (Fig. 2). D2/D4/D6 have completed training; D8/D10 and scratch D12 are not final here. No six-expert routing result is claimed.

A controlled capacity frontier. The scratch models follow the official 105-epoch recipe on all 384,795 Open Images train.0/1/2 images [9]. At fixed DIV2K epoch-20/30 milestones, 2,000 actual bitstreams per milestone show D6's RGB advantage over D2 at 0.2 payload bpp changing from 0.183 to 0.107 dB; D6 versus D4 at epoch 30 is unresolved. A separate final Kodak evaluation at 0.2 *estimated* bpp finds D4

0.116 dB [0.094, 0.139] above D2 and D6 another 0.064 dB [0.052, 0.077] above D4 on 24 paired images. Against the released D12, their image-mean YUV 6:1:1 BD-rate costs are +7.99, +4.54 and +2.79%, respectively, on four-model common support. The release has a different training history and rates are entropy estimates. The supplement gives paired uncertainty and protocol limits. Same-recipe D12 and matched shared-exit training are needed to isolate depth and weight sharing.

The spatial cost of specialisation. Patchwise expert selection also changes geometry and entropy boundaries. With identical D6 weights, independently coded regions lose quality at matched rate; increasing context alone does not remove their rate cost. In a separate fixed D2/D6 assignment, merging compatible neighbouring regions improves matched-rate quality while retaining the same per-pixel depth map. These controls, detailed in the supplement, make patchwise D12 and blind expert mixtures essential baselines before a selector receives credit. The eventual comparison must match achieved rate and quality, charge signalling and grouping, and measure the effect of expert occupancy on latency. It will test when independent specialisation is worth more than shared-prefix reuse.

6. Discussion

Spatial placement matters most at tight quality targets. Near the cheapest exit's ceiling, blind mixing approaches learned routing; validation should choose when the added decision cost pays for itself.

Source calibration reveals allocation potential, but fixed-control replay does not yet resolve its gain over blind mixing. A content-grouped unseen cohort and paired native wall-clock tests must include prediction, tile movement, entropy and fallback. Intervals here vary images at fixed weights, not training seeds. Intra YCbCr/RGB results do not establish perceptual equivalence or performance on the video path.

7. Conclusion

RegLIC makes synthesis depth a spatial decision inside DCVC-UF. Aligned exits and a shrinking tile batch reuse a common representation while omitting later blocks. Shallower depth supplies most measured arithmetic saving; learned placement adds a smaller, budget-dependent increment. Matched training and complete execution measurements will decide when independent-depth experts justify losing shared computation.

References

- [1] Théophile Blard, Pierrick Philippe, Théo Ladune, Xiaoran Jiang, and Olivier Déforges. Spatial competition for low-complexity learned image compression. In *IEEE International Conference on Image Processing (ICIP)*, 2026. Accepted; author manuscript arXiv:2605.13243. 3
- [2] Mallesham Dasari, Kumara Kahatapitiya, Samir R. Das, Aruna Balasubramanian, and Dimitris Samaras. Swift: Adaptive video streaming with layered neural codecs. In *19th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, pages 103–118. USENIX Association, 2022. 2
- [3] Zhihao Hu and Dong Xu. Complexity-guided slimmable decoder for efficient deep video compression. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14358–14367, 2023. 2
- [4] Zhaoyang Jia, Bin Li, Jiahao Li, Wenxuan Xie, Linfeng Qi, Houqiang Li, and Yan Lu. Towards Practical Real-Time Neural Video Compression. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12543–12552, 2025. 2
- [5] Jong Wook Kim, Suyong Bahk, TaeHwa Lee, HyunDong Cho, Donghyun Kim, Sung-Chang Lim, Jin Soo Choi, and Hui Yong Kim. Block-based learned image compression without blocking artifacts. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. 3
- [6] Xiangtao Kong, Hengyuan Zhao, Yu Qiao, and Chao Dong. ClassSR: A General Framework to Accelerate Super-Resolution Networks by Data Characteristic. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12016–12025, 2021. 1, 2
- [7] Jiahao Li, Wenxuan Xie, Zhaoyang Jia, Bin Li, Zongyu Guo, Xiaoyi Zhang, and Yan Lu. Ultra-Fast Neural Video Compression. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 41311–41321, 2026. 1, 2
- [8] Xintian Mao, Qingli Li, and Yan Wang. AdaRevD: Adaptive Patch Exiting Reversible Decoder Pushes the Limit of Image Deblurring. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 25681–25690, 2024. 2
- [9] Microsoft. DCVC-UF Image Training Recipe. Source repository, commit cbdae87a5445114cdc7f48816da63ea80bdeac40, 2026. 8
- [10] Calvin-Khang Ta, Praneet Singh, Tong Shao, and Peng Yin. MixCompress: Mixture of experts for variable rate learned image compression. In *European Conference on Computer Vision (ECCV)*, 2026. Accepted; author manuscript arXiv:2607.14334. 2
- [11] Lvfang Tao, Wei Gao, Ge Li, and Chenhao Zhang. Adanic: Towards practical neural image compression via dynamic transform routing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16879–16888, 2023. 1, 2
- [12] Shizun Wang, Jiaming Liu, Kaixin Chen, Xiaoqi Li, Ming Lu, and Yandong Guo. Adaptive Patch Exiting for Scalable Single Image Super-Resolution. In *European Conference on Computer Vision (ECCV)*, 2022. 1, 2
- [13] Qunliang Xing, Mai Xu, Tianyi Li, and Zhenyu Guan. Early exit or not: Resource-efficient blind quality enhancement for compressed images. In *European Conference on Computer Vision (ECCV)*, 2020. 2
- [14] Fei Yang, Luis Herranz, Yongmei Cheng, and Mikhail G. Mozerov. Slimmable Compressive Autoencoders for Practical Neural Image Compression. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2
- [15] Yibo Yang and Stephan Mandt. Computationally-Efficient Neural Image Compression with Shallow Decoders. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 2
- [16] Yufeng Zhang, Wenrui Dai, Hang Yu, Shizhan Liu, Junhui Hou, Jianguo Li, and Weiyao Lin. ABC: Adaptive bayesnet structure learning for computational scalable multi-task image compression. arXiv:2506.15228, 2025. 2
- [17] Zhichen Zhang, Bolin Chen, Hongbin Lin, Jieliang Lin, Xu Wang, and Tiesong Zhao. ELFIC: A learning-based flexible image codec with rate-distortion-complexity optimization. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 9252–9261, 2023. 2
- [18] Jiancheng Zhao, Xiang Ji, Yifan Zhan, Zunian Wan, and Yinqiang Zheng. MoECodec: Image compression for joint human and machine perception via mixture-of-experts, 2026. 2